Machine Learning-Enhanced Portfolio Optimisation for Emerging Markets

A Two-Stage Approach for the Nigerian Stock Exchange

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PhD Progress Report · 2026

Presentation Outline

1. Problem Statement
2. Progress Since the Proposal
3. Implemented Framework
4. Findings
5. Research Artefact & Live Demo
6. Next Steps
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Problem Statement

Core Research Problem

Classical portfolio methods assume normal returns, stable volatility and liquid markets; the NGX violates all three. ML portfolio frameworks are validated on developed and Asian markets (Huang et al., 2024; Orra et al., 2025), not on African frontier exchanges (Adeabah et al., 2023).

Why classical methods struggle

- **Fat tails:** 39/39 stocks non-normal (Jarque–Bera test)
- **Persistent volatility:** GARCH(1,1) persistence ≈ 0.90
- **Thin, episodic trading:** volume comes in bursts
- **Oil and naira shocks:** regime-driven volatility
- **MVO estimation error** (DeMiguel et al., 2009)

First four: NGX findings of this research (Isichei and Oduwole, 2026)

Gaps this study addresses

- No validated **two-stage GNN–RL** framework for the NGX
- Stock-network structure unexploited (Patel et al., 2024)
- RL untested under NGX costs (Ngo et al., 2023)
- Sentiment's **portfolio value** untested (John et al., 2022)
- NGX never characterised for ML design

Aim: design, validate and deploy an NGX-specific two-stage GAT–PPO portfolio framework

Abbreviations: NGX: Nigerian Exchange; ML: Machine Learning; GARCH: Generalised Autoregressive Conditional Heteroskedasticity; MVO: Mean-Variance Optimisation; GNN: Graph Neural Network; RL: Reinforcement Learning; GAT: Graph Attention Network; PPO: Proximal Policy Optimisation.

Progress at a Glance: Proposal Defense → Today

Research question	At proposal defense (Jun 2026)	Now (Oct 2026)	Status
RQ4 NGX characterisation	22 stocks, 2015–2025; paper under review	**Published**, PMLR Vol. 319; panel expanded to 39 stocks, 2012–2025	**Done**
RQ1 GAT pre-selection	Design only	GAT + 2 baselines, 7 walk-forward folds; best quintile spread (2.22%/month)	**Prototype**
RQ2 RL allocation	Design only	PPO + MVO + equal weight, 68 out-of-sample months; Sharpe 1.077	**Prototype**
RQ3 Sentiment	Conditional	1,954 headlines scored; no significant gain ($t = -0.82$, $p \approx 0.41$)	**Done**
Artefact	–	**Nairafolio** robo-advisor deployed (GAT + PPO API)	**Done**

In one sentence

The full pipeline (data → models → walk-forward evaluation → deployed advisor) now runs end to end; models are competitive with benchmarks but not yet clearly superior.

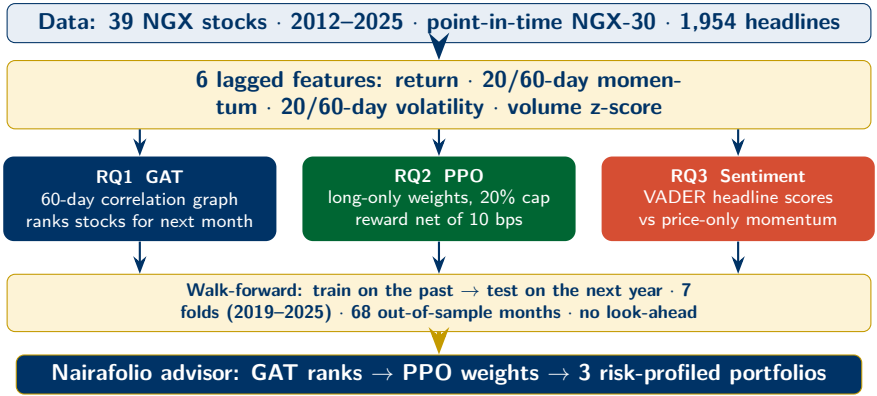
Abbreviations: RQ: Research Question; NGX: Nigerian Exchange; PMLR: Proceedings of Machine Learning Research; GAT: Graph Attention Network; RL: Reinforcement Learning; PPO: Proximal Policy Optimisation; MVO: Mean-Variance Optimisation; API: Application Programming Interface; t: t-statistic; p: p-value.

What Changed Since the Proposal, and Why

Area	Proposed	Implemented	Reason
Sample	2015–2025, 30–50 stocks	2012–2025, 39 PIT NGX-30 stocks (incl. former)	Longer; no survivorship bias
Validation	5 rolling windows	7 annual folds (2019–25), 68 test months	More out-of-sample data
GAT	3 layers, 8 heads, ranking loss	2 layers, 2 heads, MSE on rank	Prototype; small data
RQ1 baselines	LSTM, RF, XGBoost, TA	Random Forest, 60-day momentum	Prototype scope
RL	DQN + multi-reward PPO	PPO; reward = return − 10 bps	PPO suits continuous weights
Two-stage link	GAT top-K → RL	Tested separately; joined in advisor	Validate each stage first
Macro data	Model inputs	Collected; used in RQ4 only	Price-based prototype
Sentiment	BusinessDay, Nairametrics, Vanguard	Google News + GDELT (VADER)	No historical archives
Metrics	Sharpe, Sortino, Calmar, CVaR, DM	Sharpe, CAGR, max DD, turnover, IC	Others planned

Abbreviations: PIT: point-in-time; NGX-30: index of the 30 largest Nigerian Exchange stocks; GAT: Graph Attention Network; MSE: Mean Squared Error; LSTM: Long Short-Term Memory; RF: Random Forest; XGBoost: Extreme Gradient Boosting; TA: Technical Analysis; RL: Reinforcement Learning; DQN: Deep Q-Network; PPO: Proximal Policy Optimisation; bps: basis points (0.01%); top-K: K highest-ranked stocks; GDELT: Global Database of Events, Language and Tone; VADER: Valence Aware Dictionary and sEntiment Reasoner; CVaR: Conditional Value at Risk; DM: Diebold–Mariano; CAGR: Compound Annual Growth Rate; DD: Drawdown; IC: Information Coefficient.

Implemented Pipeline and Validation



GAT and PPO are evaluated separately in the experiments; joining them inside the model is the next phase.

Abbreviations: NGX-30: index of the 30 largest Nigerian Exchange stocks; RQ: Research Question; GAT: Graph Attention Network; PPO: Proximal Policy Optimisation; VADER: Valence Aware Dictionary and sEntiment Reasoner; bps: basis points (0.01%).

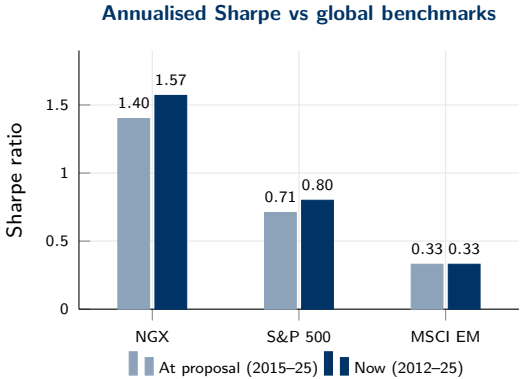
RQ4 – The NGX Is Structurally Different

Published

Five findings (now · 39 stocks)

- **Non-normal:** 39/39 fail Jarque–Bera; kurtosis 4.66
- **Persistent volatility:** GARCH $\alpha + \beta = 0.90$
- **Episodic liquidity:** 26% above 75th-pctl CoV
- **Sparse links:** mean $\rho = 0.07$; banks $\bar{\rho} = 0.25$
- **High naira Sharpe**; naira −90% vs USD

At proposal (22 stocks): kurtosis 4.79, GARCH 0.912, $\bar{\rho} = 0.109$ (Isichei and Oduwole, 2026)

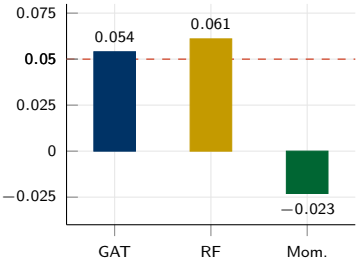


Abbreviations: RQ: Research Question; NGX: Nigerian Exchange; GARCH: Generalised Autoregressive Conditional Heteroskedasticity ($\alpha + \beta$: volatility persistence); CoV: Coefficient of Variation; pctl: percentile; ρ: Pearson correlation; USD: US dollar; S&P 500: Standard & Poor's 500 index; MSCI EM: Morgan Stanley Capital International Emerging Markets index.

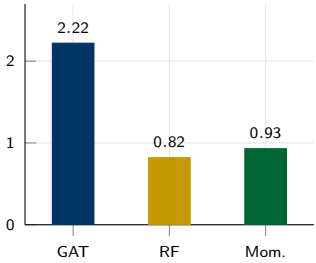
RQ1 – GAT Pre-Selection vs Baselines

Prototype

Mean IC (Spearman)



Top–bottom quintile spread (%/month)



Reading

- GAT: **widest spread**: its top picks beat its bottom picks most
- RF: slightly **higher IC**, positive in 7/7 folds (GAT 6/7)
- Momentum **fails** on the NGX (IC < 0)
- GAT median spread 1.54%; 2022 lifts the mean

Verdict

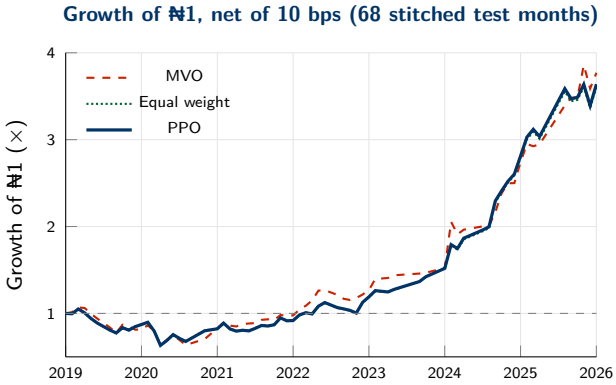
ML beats technical analysis; GAT vs RF not yet significant (1 seed).

Means of 7 folds (2019–2025); dashed line: IC = 0.05. GAT (Veličković et al., 2018); RF (Breiman, 2001); Mom. (Jegadeesh and Titman, 1993).

Abbreviations: RQ: Research Question; GAT: Graph Attention Network; RF: Random Forest; Mom.: 60-day Momentum; IC: Information Coefficient; NGX: Nigerian Exchange; ML: Machine Learning; seed: random initialisation of a model run.

RQ2 – PPO Allocation vs MVO and Equal Weight

Prototype



	PPO	MVO	EW
CAGR	25.6%	**26.4%**	25.4%
Sharpe	**1.077**	1.024	1.072
Max DD	−39.8%	−40.5%	−**39.6**%
Turnover/mo	15.2%	39.8%	**13.4%**

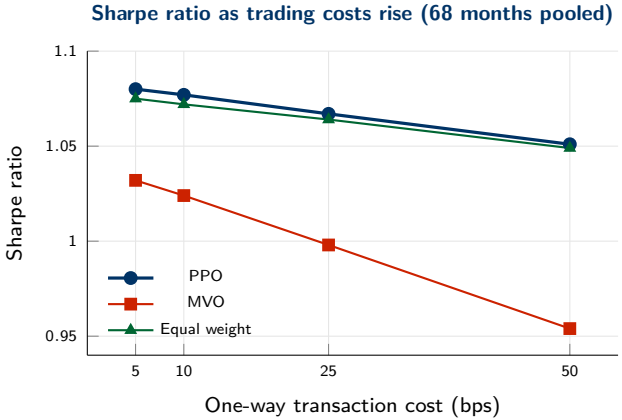
Key finding

PPO's monthly returns correlate **0.9999** with equal weight: the agent has **learned** $1/N$, a notoriously hard benchmark (DeMiguel et al., 2009).

Likely causes: 2,048 training steps per fold, return-only reward, no GAT input.

Abbreviations: RQ: Research Question; PPO: Proximal Policy Optimisation; MVO: Mean-Variance Optimisation; EW: Equal Weight; CAGR: Compound Annual Growth Rate; Max DD: Maximum Drawdown; mo: month; bps: basis points (0.01%); $1/N$: equal weighting across N stocks; GAT: Graph Attention Network; ₦: naira.

RQ2 – Where PPO Already Wins: Cost Robustness



Reading

- 5 → 50 bps: PPO loses **0.029** Sharpe; MVO loses **0.078**
- At 50 bps PPO's CAGR (24.7%) overtakes MVO's (24.1%)
- Cause: PPO trades **2.6**× **less** than MVO

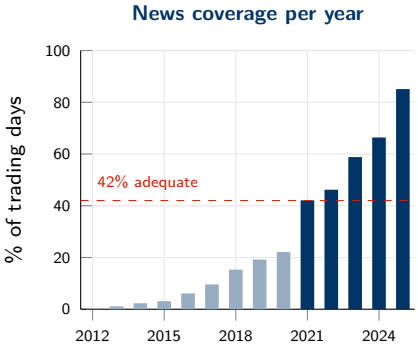
Implication for NGX investors

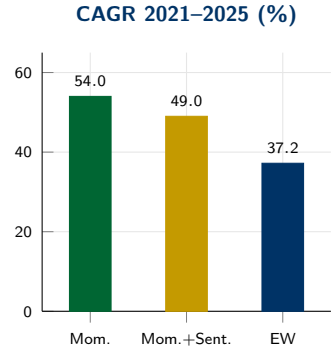
Real NGX dealing costs exceed 10 bps; low-turnover allocation keeps more of the return.

Abbreviations: RQ: Research Question; PPO: Proximal Policy Optimisation; MVO: Mean-Variance Optimisation; bps: basis points (0.01%); CAGR: Compound Annual Growth Rate; NGX: Nigerian Exchange.

RQ3 – Does News Sentiment Add Value?

Complete





Result

- Adding sentiment: $-$**3.45 pp/yr**
- 95% CI $[-11.7\%, +4.8\%]$; HAC $t = -0.82$, $p \approx 0.41$ (Newey and West, 1987)
- Full 2012–25 run agrees ($t = -0.42$); VADER scoring (Hutto and Gilbert, 2014)
- Equal weight: best Sharpe (1.639)

Caveat

Sparse news before 2021; VADER is a general-purpose lexicon.

Abbreviations: RQ: Research Question; Mom.: price-only Momentum; Sent.: Sentiment; EW: Equal Weight; CAGR: Compound Annual Growth Rate; pp: percentage points; CI: Confidence Interval; HAC: Heteroskedasticity and Autocorrelation Consistent; VADER: Valence Aware Dictionary and sEntiment Reasoner.

Which Model Performs Best?

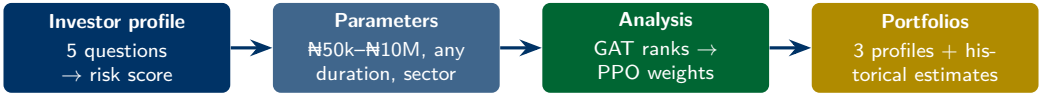
Contest	Winner	Runner-up	Verdict
Ranking accuracy (IC)	Random Forest 0.061	GAT 0.054	Close; RF more consistent
Winners vs losers (spread)	**GAT 2.22%**	Momentum 0.93%	GAT best where profit is made
Risk-adjusted return	**PPO 1.077**	Equal wt 1.072	Practically a tie
Raw return (CAGR)	MVO 26.4%	PPO 25.6%	MVO, at 2.6× the trading
Cost robustness	**PPO**	Equal weight	PPO degrades least
Sentiment increment	Price-only	+ Sentiment	No significant value

Overall

Proposed models are **competitive but not yet clearly superior**: GAT shows the strongest promise; PPO currently reproduces equal weighting.

Abbreviations: IC: Information Coefficient; RF: Random Forest; GAT: Graph Attention Network; PPO: Proximal Policy Optimisation; MVO: Mean-Variance Optimisation; CAGR: Compound Annual Growth Rate; wt: weight.

From Research to Product: the Nairafolio Robo-Advisor



Profile	Stocks	CAGR	Sharpe	Worst fall
Conservative	10	28.9%	1.12	−30.2%
Moderate	8	34.7%	1.10	−37.5%
Aggressive	5	35.4%	0.89	−38.2%
Full PPO	∼30	25.6%	1.08	−39.8%

Walk-forward replay of the saved models, 2019–2025, monthly rebalancing, 10 bps. Rule selected on the same months: estimates likely optimistic.

Built on the findings

- Liquidity filter from **RQ4**
- Risk set by concentration: fewer stocks = more risk
- Monthly rebalancing, as validated
- Next.js app + FastAPI model service

Abbreviations: GAT: Graph Attention Network; PPO: Proximal Policy Optimisation; CAGR: Compound Annual Growth Rate; RQ: Research Question; bps: basis points (0.01%); ₦: naira; k: thousand; M: million; Next.js: web-app framework; FastAPI: Python web-service framework.

Live Demo

Nairafolio: NGX Robo-Advisor

`nairafolio-nine.vercel.app`

1. Overview: headline results
2. Research: interactive figures
3. Advisor: ₦750,000 · 8 months · Oil & Gas

Scan to try it yourself

Next Phase: Closing the Gaps and Sharpening Novelty

RQ4 design requirements vs prototype

Downside (drawdown/CVaR) reward	**Not yet**
GARCH / volatility-regime features	**Partial**
Monthly rebalancing, NGX costs	**Partial**
Attention checked on bank cluster	**Partial**
Outlier exclusion (TRANSCORP)	**Not yet**

Target novelty

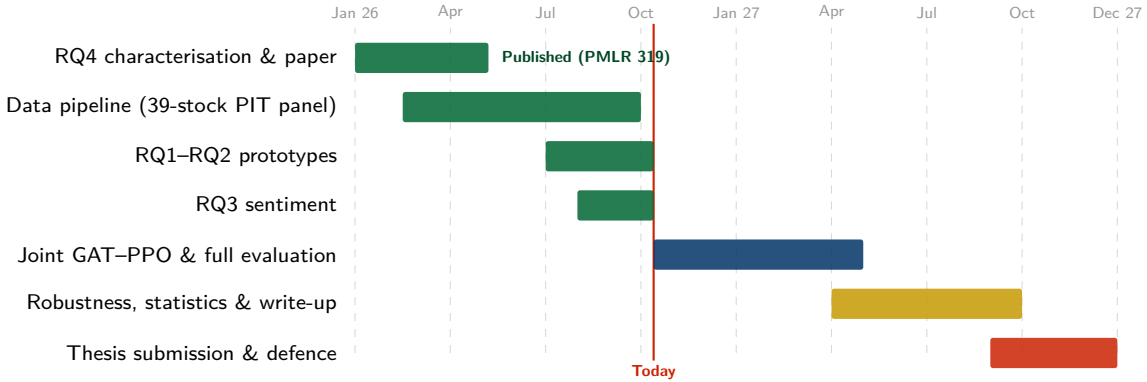
Liquidity- and macro-aware GAT informing a downside-risk PPO, validated walk-forward on a survivorship-free NGX panel.

Prioritised improvements

1. **Join the stages:** GAT scores in PPO's state (Huang et al., 2024)
2. **Fix PPO:** 50k+ steps, downside reward, 5–10 seeds
3. **NGX graph:** sector + liquidity + oil/FX context node; ranking loss
4. **GAT + RF stack** to combine strengths
5. **Statistics:** DM (Diebold and Mariano, 1995), Sharpe test (Ledoit and Wolf, 2008); Sortino, Calmar, CVaR
6. **Data:** total returns; USD and real-terms results
7. **RQ3+:** FinBERT; sentiment as a GAT node feature

Abbreviations: RQ: Research Question; CVaR: Conditional Value at Risk; GARCH: Generalised Autoregressive Conditional Heteroskedasticity; NGX: Nigerian Exchange; TRANSCORP: Transnational Corporation Plc; GAT: Graph Attention Network; PPO: Proximal Policy Optimisation; RF: Random Forest; FX: foreign exchange; DM: Diebold–Mariano test; USD: US dollar; FinBERT: Financial Bidirectional Encoder Representations from Transformers; k: thousand.

Revised Timeline



Green: completed · Navy/Gold/Red: remaining phases. Original plan: 24 months from January 2026.

Abbreviations: RQ: Research Question; PIT: point-in-time; GAT: Graph Attention Network; PPO: Proximal Policy Optimisation; PMLR: Proceedings of Machine Learning Research.

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Thank You

Questions & Discussion

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